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Introduction

CMMI (Capability Maturity Model) is a framework used to improve an organization's processes, including software development, project management, and quality assurance.

The maturity levels describe how well an organization manages and improves its processes.

CMMI Level 1 (Initial)

Process Areas at Level 1

There are no specific Process Area (PAs) assigned to CMMI Level 1. Organizations at this level work in an ad-hoc and unpredictable manner. Success mainly depends on individual effort rather than defined organizational processes.

How IT Companies Work at Level 1

Developed Countries

- Small startups may initially operate this way
- Processes are informal
- Documentation is limited
- Success depends on experienced team members.

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Non-developed / Developing Countries

Many small software houses and freelance-based organizations work at this level.
Requirements change frequently
Few formal quality controls exist
Project outcomes depend heavily on individuals.

CMMI Level 2 (Managed)

At Level 2, projects are planned, monitored, measured, controlled
Basic project management practices are established.

Process Areas of Level 2

1. **REQM** : Requirements Management
2. **PP** : Project Planning
3. **PMC** : Project Monitoring and Control
4. **MA** : Measurement and Analysis
5. **PPQA** : Process and Product Quality Assurance
6. **CM** : Configuration Management
7. **SAM** : Supplier Agreement Management

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Requirements Management (REQM)

Developed-country companies use tools like Jira, Azure DevOps to document and track requirements.

Developing-country companies use client meetings, spreadsheets and manual tracking.

Project Planning (PP)

Developed-country companies use standardized methods and project management software for planning.

Developing-country companies often rely on basic tools and managers' experience to prepare schedules, budgets, and resource plans.

Project Monitoring and Control (PMC)

Developed-country companies monitor progress using dashboards, performance metrics and automated reports.

Developing-country companies often use status meetings, progress reports, and manual tracking.

Measurement and Analysis (MA)

Developed-country companies analyze detailed metrices, such as productivity and defect rates.

Developing-country companies often focus on basic project indicators and simpler reports.

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Process and Product Quality Assurance

Developed-country companies use dedicated QA teams for audits, reviews, and compilence checks.

Developing-country companies often rely on testing, code reviews, and periodic quality inspections.

Configuration Management (CM)

Developed-country companies use tools like Git and automated version control.

Developing-country companies also use version control, but may rely more on manual controls and team discipline.

Supplier Agreement Management (SAM)

Developed-country companies manage suppliers through formal contracts, SLAs, and performance reviews.

Developing-country companies often focus on contracts and basic monitoring of outsourced services.

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CMMI Level 3 (Defined)

At Level 3, processes are not only managed but also standardized across the entire organization. All teams follow documented organizational processes.

Process Area of Level 3

RD	Requirements Development
TS	Technical Solution
PI	Product Integration
VER	Verification
VAL	Validation
RSKM	Risk Management
DAR	Decision Analysis and Resolution
IPM	Integrated Project Management
OPD	(Organizational) Process Definition
OPF	(Organizational) Process Focus
OT	Organizational Training

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RD

Developed-country companies use formal methods to gather, analyze and document requirements.

Developing-country companies often rely on client meetings, interviews and basic documentation to define requirements.

TS

Developed-country companies use standard design methods and advanced tools to develop software solutions.

Developing-country companies often use simpler design tools and rely more on developers experience.

PI

Developed-country companies use automated tools and defined procedures to combine software components.

Developing-country companies often integrate components through manual testing and basic integration procedures.

VER

Developed-country companies use systematic reviews, inspections, and automated testing to check whether software meets specified requirements.

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Developing-country companies often rely on manual testing and code reviews.

VAL

Developed-country companies use user acceptance testing and structured feedback to ensure software meets user needs.
Developing-country companies often use client feedback, demonstrations and manual acceptance testing.

RSKM

Developed-country companies identify, assess and monitor risks using formal risk management plans and tools
Developing-country companies often manage risks through team discussions on basic risk registers.

DAR

Developed-country companies compare alternatives using defined criteria, data, and documented analysis.
Developing-country companies often make decisions through team discussions and managers experience.

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IPM

Developed-country companies coordinate projects using standardized processes, shared plans and collaboration tools.
Developing-country companies often use coordinate teams through meetings, project plans, and basic management tools.

OPD

Developed-country companies establish standard procedures, templates and guidelines for software development.
Developing-country companies often use basic documentation and reusable templates to maintain consistency.

OPF

Developed-country companies regularly evaluate processes and plan improvements using performance data and feedback.
Developing-country companies often identify weaknesses through project reviews and team feedback.

OT

Developed-country companies provide structured training programs to improve employees technical and professional skills.

Developing-country companies often use workshops, online courses, and on-the-job training